

Phase 8: Data Management & Deployment

1. Data Import Wizard

- The **Data Import Wizard** is Salesforce's point-and-click tool for importing simple data into standard and custom objects.

◆ How we applied it in our project:

- Imported candidate details (`Candidate__c`) such as Name, Email, Phone, and Address.
- Bulk uploaded **Skills** and **Certifications** related to candidates for initial setup.
- Mapped columns in CSV files to Salesforce fields to ensure data accuracy.

Key Points:

- Supports up to **50,000 records** at a time.
- Provides field mapping for custom objects (like our **Candidate__c**, **Skill__c**, and **Job__c**).
- Suitable for **administrators** with minimal technical skills.

(Import candidate, skill & job data)

1. Go to **Setup** → **Data Import Wizard**.
2. Select the object:
 - `Candidate__c` (custom object).
 - `Skill__c` (skills for candidates).
 - `Job__c` (job openings).
3. Click **Launch Wizard**.
4. Upload CSV file (example: `candidates.csv`).

```
FirstName,LastName,Email,Phone,Location
Rupa,Sree,rupa@example.com,9876543210,Hyderabad
Arjun,Kumar,arjun@example.com,9876500000,Bangalore
```

Sample csv files =>

```
Candidate_Name__c,Student_Name__c,Email__c,ContactNumber__c,Education__c,Place__c,Skills__c
Rupa Sree,STU-001,rupa@example.com,9876543210,Graduate,Hyderabad,Java;
Salesforce; Apex
Arjun Kumar,STU-002,arjun@example.com,9876500000,Diploma,Bangalore,Python;
Data Analysis
Meena Devi,STU-003,meena@example.com,9876511111,Post Graduate,Chennai,Cloud
Computing; AI
```

Job_Name__c,Job_Title__c,Job_Department__c,Job_Description__c,Location__c,Locationn__c,Required_Experience__c,salary_range__c,Application_Status__c
 J-001,Software Engineer,IT,Develop and maintain applications,17.3850,-78.4867,2,6 LPA,Open
 J-002,Data Analyst,Analytics,Analyze data and prepare dashboards,12.9716,77.5946,1,4.5 LPA,Open
 J-003,HR Executive,HR,Manage recruitment and employee relations,28.7041,77.1025,0,3 LPA,Closed

Name,Candidate__c,Job__c,Application_Status__c
 APP-001,CAN-001,J-001,Submitted
 APP-002,CAN-002,J-002,Under Review
 APP-003,CAN-003,J-003,Rejected

Getting closer...

Choose data Edit mapping Start import

Import your Data into Salesforce

You can import up to 50,000 records at a time.

[Help for this page](#)

What kind of data are you importing?

Standard objects

Custom objects

- Applications
- Candidates**
- Departments
- Error_Logs
- Expense Items
- Jobs
- Skills
- Students

What do you want to do?

Add new records

Update existing records

Add new and update existing records

Match by: **Name**

Which User field in your file designates record owners? **--None--**

Which Student field in your file specifies the Master/Detail relationship? **Salesforce.com ID**

Trigger workflow rules and processes?

☐ Trigger workflow rules and processes for new and updated records

Record type **Fresher Candidate**

Where is your data located?

Drag CSV file here to upload

CSV

File

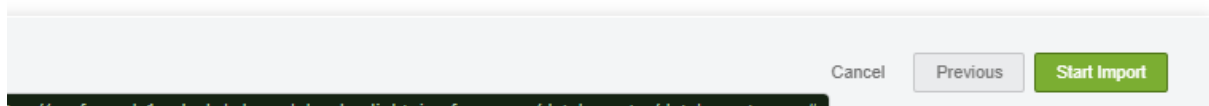
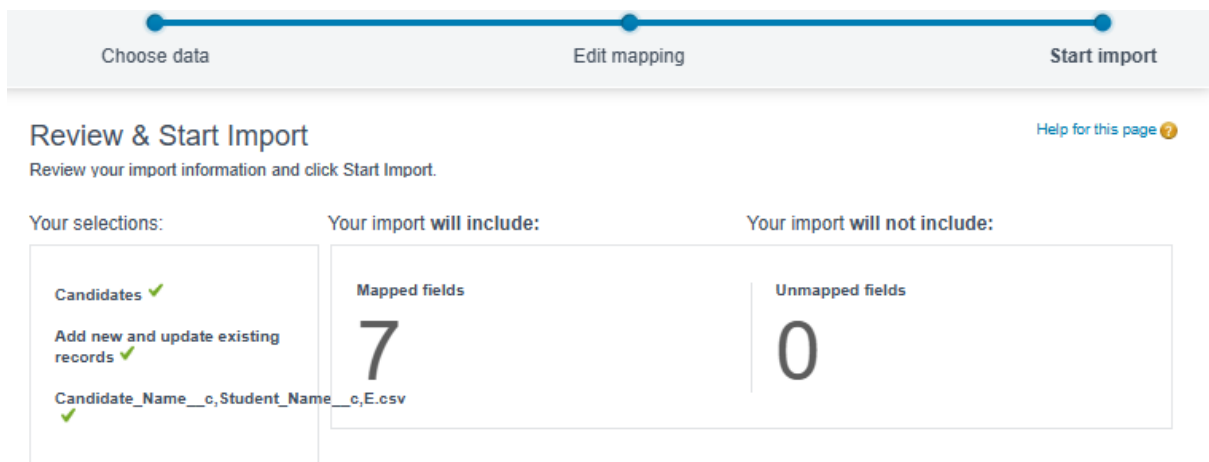
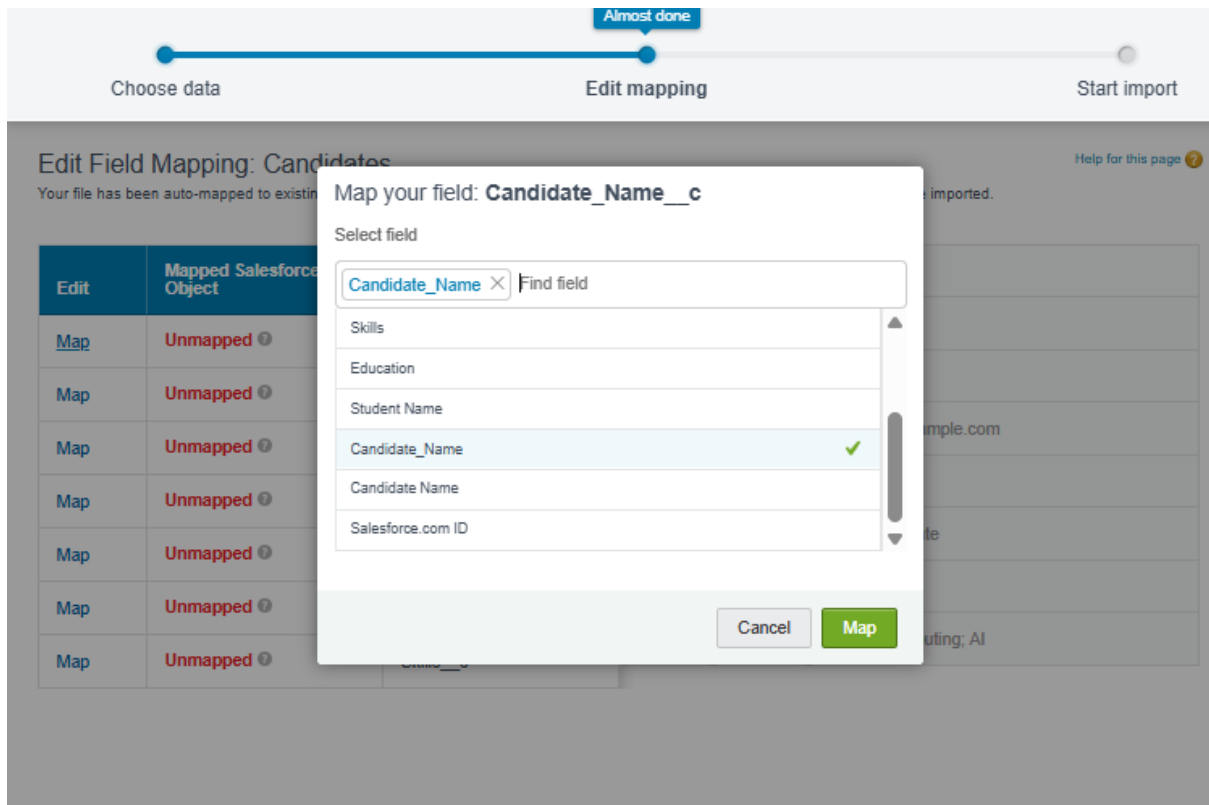
Choose File Candidate_...me__c.E.csv

Character Code **ISO-8859-1 (General US & Western European, I)**

Values Separated By **Comma**

Cancel Previous **Next**

5. Map fields → CSV columns → Salesforce fields.



6. Start Import.

7. Monitor progress in **Bulk Data Load Jobs**.
For Candidate Object=>

Object Manager

 **SETUP**
Bulk Data Load Jobs

Bulk Data Load Job
750gL00000E0yvN

Help for this Page ?

View the details of a bulk data load job.

< [Back to List: Bulk Data Load Jobs](#)

Bulk Data Load Job Detail

Reload

Job ID	750gL00000E0yvN	Job Type	Bulk V1	Status	Closed
Submitted By	T. Rupasree	Operation	Upsert	Total Processing Time (ms)	76
Start Time	9/23/2025, 8:23 PM PST	Queued Batches	0	API Active Processing Time (ms)	29
End Time	9/23/2025, 8:23 PM PST	In Progress Batches	0	Apex Processing Time (ms)	0
Time to Complete ([hh:mm:ss])	00:01	Completed Batches	1		
Object	Candidate	Failed Batches	0		
External ID Field	Name	Progress	100%		
Content Type	CSV	Records Processed	3		
Concurrency Mode	Parallel	Records Failed	3		
API Version	64.0	Retries	0		

Reload

Batches

View Request	View Result	Batch ID	Start Time	End Time	Total Processing Time (ms)	API Active Processing Time (ms)	Apex Processing Time (ms)	Records Processed	Records Failed
View Request	View Result	751gL00000BaOAG	9/23/2025, 8:23 PM	9/23/2025, 8:23 PM	76	29	0	3	3

For Job Object=>



SETUP

Bulk Data Load Jobs

Bulk Data Load Job

750gL00000E0zhl

[Help for this Page](#) ?

View the details of a bulk data load job.

[← Back to List: Bulk Data Load Jobs](#)

Bulk Data Load Job Detail

Reload

Job ID	750gL00000E0zhl	Job Type	Bulk V1	Status	Closed
Submitted By	T. Rupasree	Operation	Upsert	Total Processing Time (ms)	108
Start Time	9/23/2025, 8:28 PM PST	Queued Batches	0	API Active Processing Time (ms)	44
End Time	9/23/2025, 8:28 PM PST	In Progress Batches	0	Apex Processing Time (ms)	0
Time to Complete ([hh:mm:ss])	00:00	Completed Batches	1		
Object	Job	Failed Batches	0		
External ID Field	Name	Progress	100%		
Content Type	CSV	Records Processed	3		
Concurrency Mode	Parallel	Records Failed	3		
API Version	64.0	Retries	0		

Reload

Batches

View Request	View Result	Batch ID	Start Time	End Time	Total Processing Time (ms)	API Active Processing Time (ms)	Apex Processing Time (ms)	Records Processed	Records Failed
View Request	View Result	751gL00000Bacus	9/23/2025, 8:28 PM	9/23/2025, 8:28 PM	108	44	0	3	3

Application object=>



SETUP

Bulk Data Load Jobs

Bulk Data Load Job

750gL00000E0wQY

[Help for this Page](#)

View the details of a bulk data load job.

[Back to List: Bulk Data Load Jobs](#)

Bulk Data Load Job Detail

[Reload](#)

Job ID	750gL00000E0wQY	Job Type	Bulk V1	Status	Closed
Submitted By	T. Rupasree	Operation	Upsert	Total Processing Time (ms)	188
Start Time	9/23/2025, 8:33 PM PST	Queued Batches	0	API Active Processing Time (ms)	98
End Time	9/23/2025, 8:33 PM PST	In Progress Batches	0	Apex Processing Time (ms)	0
Time to Complete (hh:mm:ss)	00:04	Completed Batches	1		
Object	Application	Failed Batches	0		
External ID Field	Name	Progress	100%		
Content Type	CSV	Records Processed	3		
Concurrency Mode	Parallel	Records Failed	0		
API Version	64.0	Retries	0		

[Reload](#)

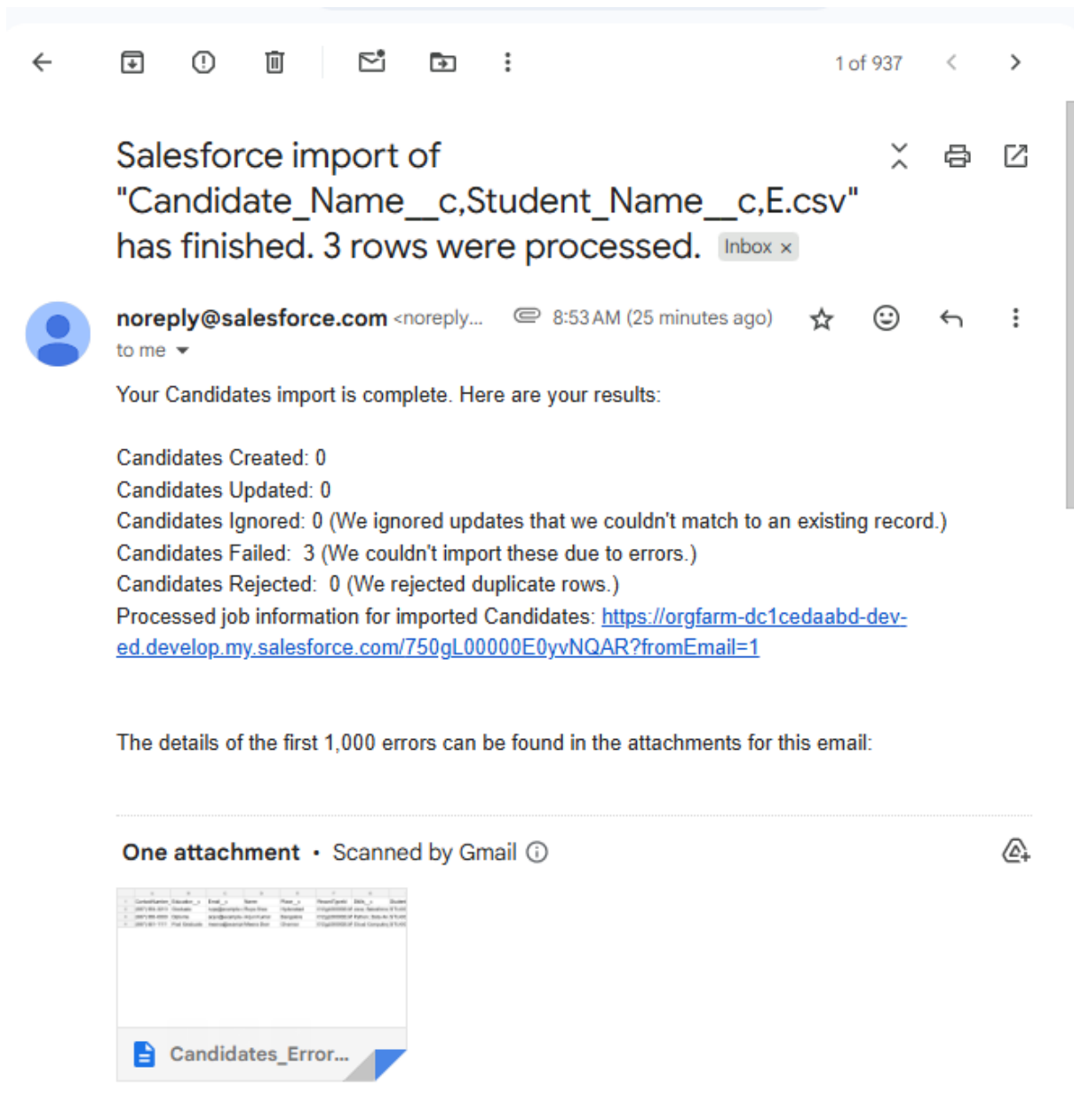
Batches

View Request	View Result	Batch ID	Start Time	End Time	Total Processing Time (ms)	API Active Processing Time (ms)	Apex Processing Time (ms)	Records Processed	Records Failed
View Request	View Result	751gL00000BaOIJ	9/23/2025, 8:33 PM	9/23/2025, 8:33 PM	188	98	0	3	0

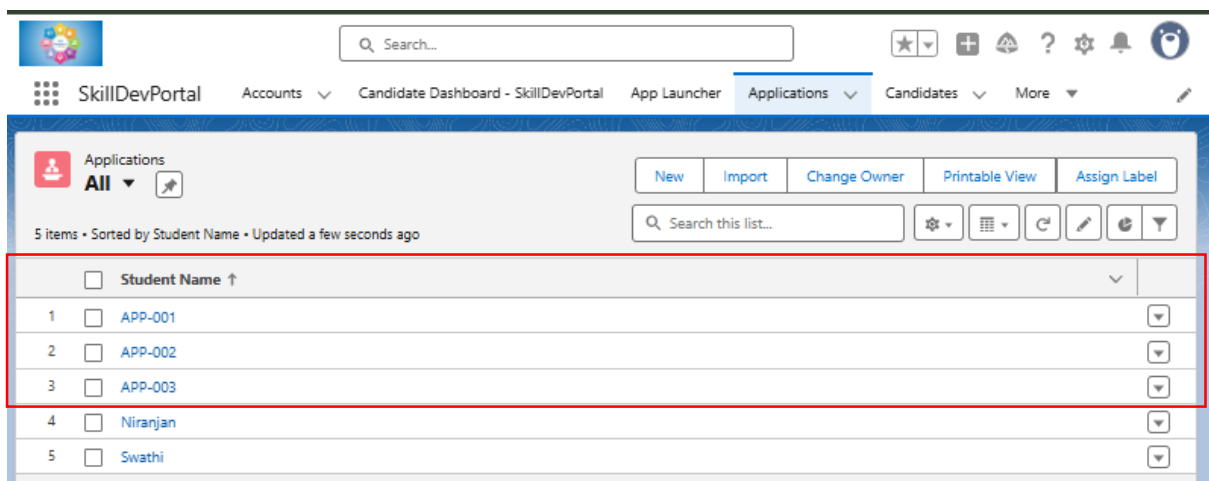
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Id	Success	Created	Error												
	FALSE	TRUE	INVALID_OR_NULL_FOR_RESTRICTED_PICKLIST:Locationn: bad value for restricted picklist field: -78.4867:Locationn__c--												
	FALSE	TRUE	INVALID_OR_NULL_FOR_RESTRICTED_PICKLIST:Job Department: bad value for restricted picklist field: Analytics:Job_Department__c--												
	FALSE	TRUE	INVALID_OR_NULL_FOR_RESTRICTED_PICKLIST:Locationn: bad value for restricted picklist field: 77.1025:Locationn__c--												

☞ In our portal: This is used for **initial candidate data load** and **bulk skill uploads** from Excel sheets given by admin.

We even get a mail after a successful import=>



Here we can see the records updated in our application object directly through csv file import=>



2. Data Loader

- The **Data Loader** is a client application for bulk importing, updating, deleting, and exporting records.

◆ How we applied it in our project:

- Imported **large datasets** of candidates and job openings.
- Used **Export** functionality to generate reports of candidate-job matching data for backup.
- Updated multiple records for skills and job assignments during testing.

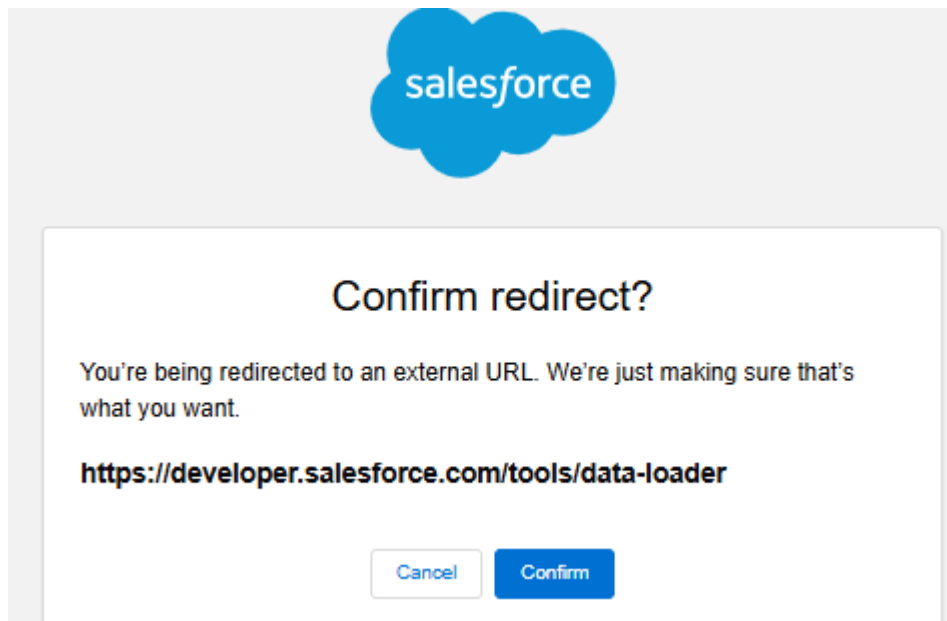
Data Loader is a **client application** (separate from Salesforce UI) used for bulk data operations. Unlike Data Import Wizard (limit: 50k records), **Data Loader supports up to 5 million records**.

When to Use Data Loader

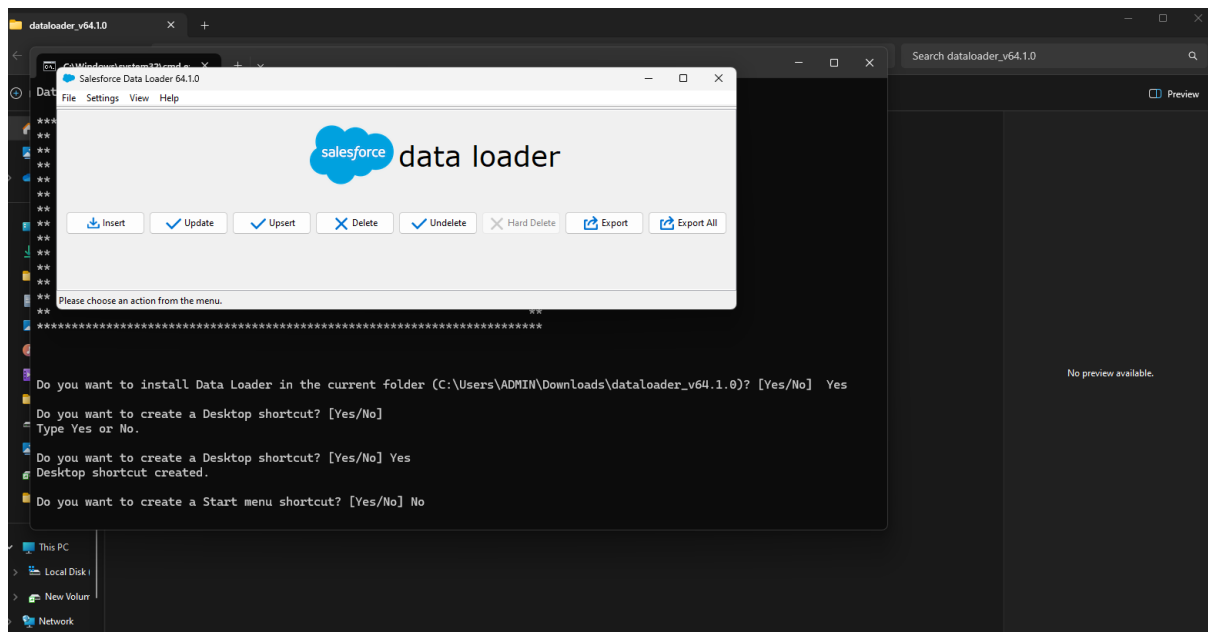
- Importing **large volumes** of data (e.g., all Candidates, Jobs, Skills at once).
 - Performing **complex mappings** or scheduled imports/exports.
 - Updating or deleting **specific record sets** using IDs.
 - Exporting data for **backup** or **migration**.
-

Steps to Use Data Loader

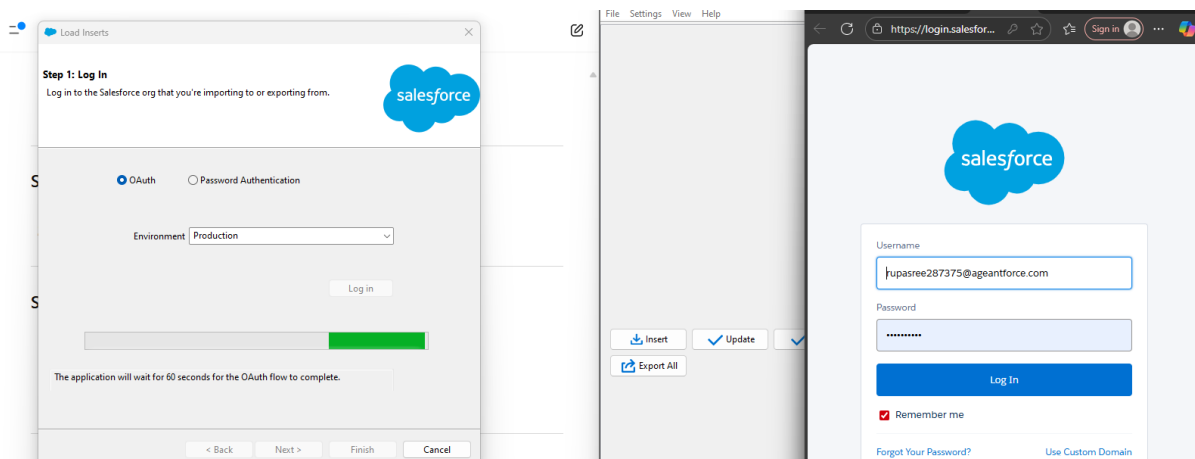
1. **Download & Install**
 - Go to **Setup** → **Data Loader** (search in Quick Find).



- Download the version for **Windows or Mac**.
- Install it on your system.



- Choose environment:
- **Production** (login.salesforce.com)



Authorization Successful!

You can now close this browser window and
return to Data Loader.

3. Choose Operation

- Options:
- **Insert** → Add new records.
- **Update** → Modify existing records.
- **Upsert** → Insert new + Update existing.
- **Delete** → Remove records.
- **Export/Export All** → Download data for backup.

4. Select Object

- Example: **Candidate__c**
- Browse and upload your **CSV file**.

Load Inserts

Step 2: Select Salesforce object

Import batch size: 200 Start at row: 0
Current API usage for the org: 1
API Limit for the org: 15,000
SOAP API version: 64.0

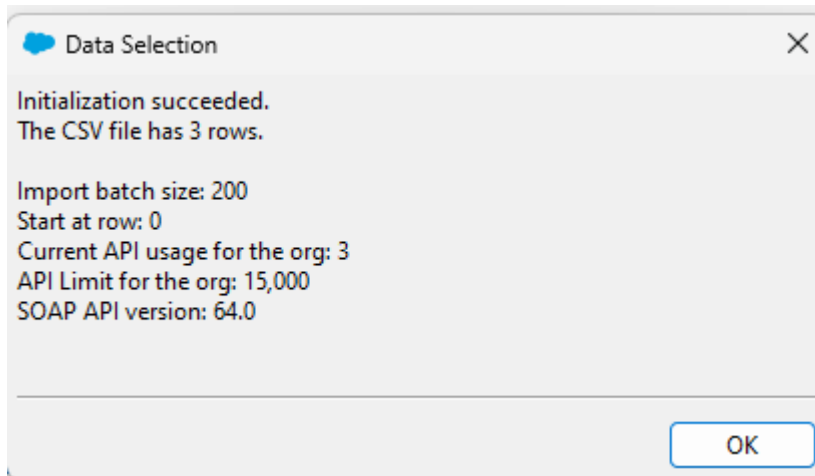
Select Salesforce object to import:

Account (Account)
Application (Application__c)
Candidate (Candidate__c)
Case (Case)
Contact (Contact)
Department (Department__c)
enrollment (enrollment__c)
Error_Log (Error_Log__c)
Event (Event)
Expense Item (Expense_Item__c)
Job (Job__c)
Lead (Lead)
mentor (mentor__c)
Opportunity (Opportunity)

☐ Show all Salesforce objects

Import from (CSV file): C:\Users\ADMIN\Downloads\Candidate_Name__c,Student_N Browse...

< Back Next > Finish Cancel



Step 2b: (Optional) relate using lookup field

Relationships of Candidate__c are listed below.
Select a related object and its lookup field if the CSV refers to the related object using the selected lookup field.



Relationship	Related Object	-	Lookup Field of Related Object
RecordType :	RecordType	-	Id
Student_Name__r :	Student__c	-	Id

5. Map Fields

- Data Loader will auto-map based on headers.
- Unmapped fields → manually drag & drop.
- Save the mapping for reuse.

Step 3: Mapping

Import batch size: 200 Start at row: 0
Current API usage for the org: 3
API Limit for the org: 15,000
SOAP API version: 64.0



Choose an Existing Map

Create or Edit a Map

CSV Column Header	Salesforce Object Field Name
Candidate_Name__c	Candidate_Name__c
Student_Name__c	Student_Name__c
Email__c	Email__c
ContactNumber__c	ContactNumber__c
Education__c	Education__c
Place__c	Place__c
Skills__c	Skills__c

< Back

Next >

Finish

Cancel

6. Run the Job

- Choose a folder for **Success.csv** & **Error.csv** logs.
- Click **Next** → **Finish**.

Step 4: Finish

Import batch size: 200 Start at row: 0
Current API usage for the org: 3
API Limit for the org: 15,000
SOAP API version: 64.0



Results Folder:

[Browse...](#)

[< Back](#)

[Next >](#)

[Finish](#)

[Cancel](#)

- Data Loader will process all rows and give logs.

Key Points:

- Supports **millions of records**.
 - Can schedule data loads for automation.
 - Allows **upsert** operations (update + insert).
-

3. Duplicate Rules

- To maintain clean candidate and employer data, **Duplicate Rules** were defined.
 - Duplicate Rules help **prevent or alert users about duplicate records** in Salesforce. This is especially important for your **Skill Development & Employment Portal**, so you don't accidentally have multiple records for the same candidate, job, or student.
-

Step 1: Go to Duplicate Rules

1. In **Salesforce Setup** → **Quick Find** → type **Duplicate Rules**.
2. Click **Duplicate Rules** under **Data** → **Duplicate Management**.
3. You'll see existing rules (e.g., Lead, Account, Contact).



SETUP

Duplicate Rules

All Duplicate Rules

[Help for this Page](#) ?

What Are Duplicate Rules?

[\[Expand \]](#)View: All Duplicate Rules ▼A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other All

New Rule ▼						
Rule Name ↕	Description	Object	Matching Rule	Active	Last Modified By	Last Modified Date
Standard Account Duplicate Rule	Identify accounts that duplicate other accounts.	Account	Standard Account Matching Rule	✓	eexec	7/19/2025
Standard Contact Duplicate Rule	Identify contacts that duplicate other contacts and leads.	Contact	Standard Contact Matching Rule Standard Lead Matching Rule	✓	eexec	7/19/2025
Standard Lead Duplicate Rule	Identify leads that duplicate other leads and contacts.	Lead	Standard Contact Matching Rule Standard Lead Matching Rule	✓	eexec	7/19/2025

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other All

Step 2: Create a New Duplicate Rule

1. Click **New Rule**.
2. Select **Object**:
 - Example: **Candidate** (Candidate__c)
3. Fill in basic info:
 - **Rule Name**: Candidate Duplicate Rule
 - **Description**: "Prevents duplicate candidate records based on Email or Contact Number."
 - **Record-Level Security**: Choose who the rule applies to (All Users / Admins Only).
4. **Activation**: Don't activate yet until testing.

Duplicate Rule Edit

Save

Save & New

Cancel

Rule Details

Required Information

Rule Name

Description

Object

Record-Level Security ☒ Enforce sharing rules ☐ Bypass sharing rules

Actions

Specify what happens when a user tries to save a duplicate record.

Action On Create ☒ Alert ☒ Report

Action On Edit ☐ Alert ☐ Report

Alert Text

Matching Rules

Define how duplicate records are identified.

Compare Candidates With

Matching Rule

Matching Criteria

Field Mapping

Add Rule

Remove Rule

Conditions

Optionally, specify the conditions a record must meet for the rule to run.

Field	Operator	Value
--None--	--None--	
--None--	--None--	
--None--	--None--	

Step 3: Define Matching Rule

- Duplicate Rules use **Matching Rules** to decide if two records are duplicates.
- Click **Create New Matching Rule** (or use existing).
 - Select **Object**: `Candidate__c`
 - Matching Criteria:
 - `Email__c` → Exact Match
 - `ContactNumber__c` → Exact Match

Step 2: Configure Matching Rule

Step 2 of 2

[Previous](#) [Save](#) [Cancel](#)

Rule Details

***** = Required Information

Object	Candidate
Rule Name	Candidate Duplicate Rule n
Unique Name	Candidate_Duplicate_Rule_ i
Description	

Matching Criteria

Tell the rule which fields to compare and how.

Field	Matching Method i	Match Blank Fields i	
Email	Exact	☐	AND
ContactNumber	Exact	☐	AND
--None--	Exact	☐	AND
--None--	Exact	☐	AND
--None--	Exact	☐	

[Add Filter Logic...](#)

[Previous](#) [Save](#) [Cancel](#)

4. Save & **Activate** the matching rule.

Matching Rule Help for this Page ?

Candidate Duplicate Rule matching rule

Matching Rule Detail Edit Delete Clone Activate

Object	Candidate
Rule Name	Candidate Duplicate Rule matching rule
Unique Name	Candidate_Duplicate_Rule_matching_rule
Description	
Matching Criteria	(Candidate: Email EXACT MatchBlank = FALSE) AND (Candidate: ContactNumber EXACT MatchBlank = FALSE)
Status	Inactive
Created By	T. Rupasree, 9/23/2025, 11:03 PM
Modified By	T. Rupasree, 9/23/2025, 11:03 PM

Matching Rule Activation

 We're activating your matching rule. We'll send an email to t.rupasree287@gmail.com when the activation process is complete.

OK

Step 4: Complete Duplicate Rule

1. Back in the Duplicate Rule setup:
 - **Actions on Create / Edit:**
 - **Block** → Prevent saving a duplicate
 - **Alert** → Warn the user but allow save
 - Example: Choose **Alert** so users are warned about duplicate candidates.
2. **Optional:**
 - Report on duplicate records found

- Email notifications to admins
3. Save the rule.



SETUP

Duplicate Rules

Edit Duplicate Rule

Candidate Duplicate Rule

Help for this Page ?

Duplicate Rule Edit

Save Save & New Cancel

Rule Details

Rule Name

Candidate Duplicate Rule

Description

Prevents duplicate candidate records based on Email or Contact Number.

Object

Candidate

Record-Level Security

☒ Enforce sharing rules
 ☐ Bypass sharing rules

Actions

Specify what happens when a user tries to save a duplicate record.

Action On Create

Block

Action On Edit

Block

Alert Text

This Email or Contact Number is already registered

Matching Rules

Define how duplicate records are identified.

Compare Candidates With

Candidates

Matching Rule

Candidate Duplicate Rule matching rule

Matching Criteria

(Candidate: Email EXACT MatchBlank = FALSE) AND (Candidate: ContactNumber EXACT MatchBlank = FALSE)

Field Mapping

Mapping Selected

Add Rule

Remove Rule

Conditions

Optionally, specify the conditions a record must meet for the rule to run.

Field	Operator	Value
--None--	--None--	
--None--	--None--	

Step 5: Test the Rule

1. Go to **Candidate Tab** → **Create New Candidate**.
2. Enter a candidate with an **existing Email or Contact Number**.
3. Salesforce will **warn you** (or block) according to your rule.

New Candidate: Fresher Candidate

* = Required Information

Information

Candidate Name

CandidateID

* Email

ContactNumber

Place

Skills

Education

Candidate Name

* Student

ⓘ We hit a snag.

Review the errors on this page.

- duplicate value found: Email__c duplicates value on record with id: a02gL0000071rELQAY

ⓘ

➔ The above screen shows that entering email or contact number that already exists raised an error to review and didn't allow to save due to our duplicate rule on candidate object

◆ **How we applied it in our project:**

- Created a rule on **Candidate__c** to prevent duplicate entries by **Email ID**.
- Configured alerts for recruiters when attempting to add duplicate job openings.

Key Points:

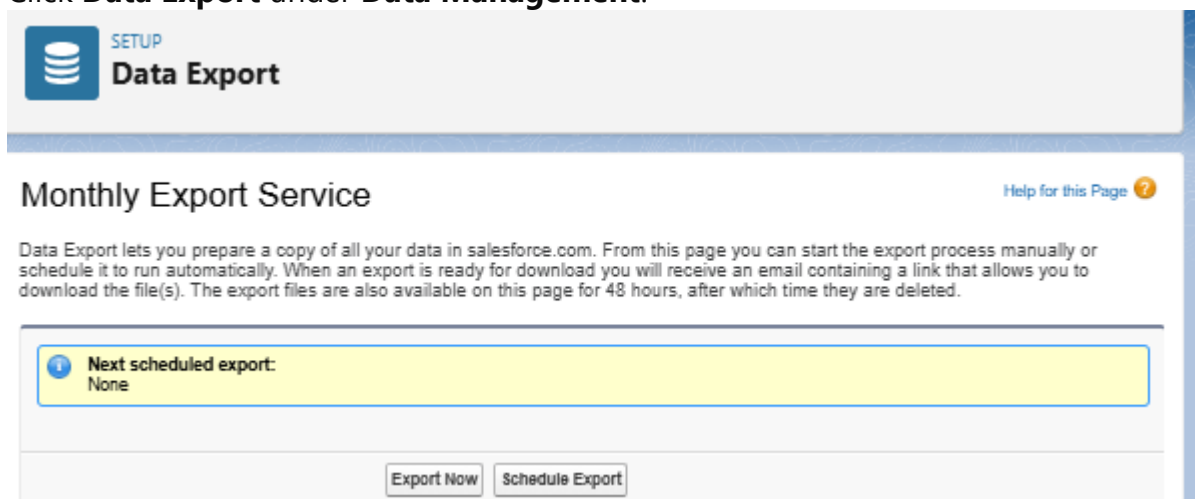
- Prevents bad data quality.
- Can be set to **Block** or **Allow with Alert**.
- Uses **Matching Rules** behind the scenes.

4. Data Export & Backup

- Data safety was ensured by enabling **Data Export Service** for backups.
- Data Export & Backup ensures your Salesforce data is **safe, recoverable, and portable**. For your **Skill Development & Employment Portal**, this is crucial to protect **Candidates, Jobs, Skills, and Applications** data.

Step 1: Go to Data Export

1. In **Salesforce Setup** → **Quick Find**, type **Data Export**.
2. Click **Data Export** under **Data Management**.



Step 2: Choose Export Type

1. **Manual Export** – one-time export.
2. **Schedule Export** – recurring weekly or monthly backups.

For initial backup, select **Manual Export**.

Step 3: Select Data

1. You'll see a list of all objects:
 - **Candidates, Jobs, Applications, Skills, Students**, etc.
2. Check the objects you want to export.
 - For example: **Candidate__c, Job__c, Application__c, Skill__c**
3. Choose options:
 - **Include all data**
 - **Include attachments** (if needed)
 - **Include Salesforce Files** (optional)

The screenshot shows the 'Monthly Export Service' interface. At the top, there's a title 'Monthly Export Service' and a 'Help for this Page' link. Below the title is a form with several options: 'Export File Encoding' set to 'Unicode (UTF-8)', 'Include Images, documents, and attachments' checked, 'Include Salesforce Files and Salesforce CRM Content document versions' checked, and 'Replace carriage returns with spaces' checked. At the bottom of the form are 'Start Export' and 'Cancel' buttons. Below the form, there's a section labeled 'Exported Data'.

Step 4: Start Export

1. Click **Start Export**.
2. Salesforce will prepare a **ZIP file** with your CSV files for each object.
3. Once ready, Salesforce will **email you a download link**.

Large data may take a few minutes to hours depending on volume.

download the file(s). The export files are also available on this page for 48 hours, after which time they are deleted.

The screenshot shows the export status and scheduling options. At the top, there's a yellow box with a blue 'i' icon and the text: 'Next scheduled export: A data export is currently in progress for your organization.' Below this are 'Export Now' and 'Schedule Export' buttons. At the bottom, there's a table with the following information:

Your export has been queued. You will receive an email notification when it is completed.	
Scheduled By	T. Rupasree
Schedule Date	9/23/2025
Export File Encoding	Unicode (UTF-8)

Step 5: Download & Store

1. Download the **ZIP file** from the link in your email.
 2. Store it in a **safe location** (local drive or cloud storage).
 3. You can now **open CSVs** for backup, offline analysis, or re-import if needed.
-

Step 6: Optional – Schedule Recurring Backups

- Click **Schedule Export** → set **frequency** (Weekly / Monthly).
- Salesforce will automatically send the export ZIP to your email.

◆ **How we applied it in our project:**

- Exported **weekly backups** of all candidate, job, and employer data.
- Downloaded CSV files and stored securely for recovery.

Key Points:

- Provides data in CSV format.
 - Can be scheduled (weekly/monthly).
 - Essential for **disaster recovery**.
-

5. Change Sets

- **Change Sets** were used to migrate customizations between environments.
 - Change Sets allow you to **deploy customizations** from one Salesforce org to another (e.g., from **Sandbox** → **Production**). This is crucial for your **Skill Development & Employment Portal** when moving objects, fields, or Lightning components after testing.
-

"The Change Sets process is being skipped due to the unavailability of a Sandbox org login and will be implemented in a subsequent next processes."

Key Points:

- Works only between **connected Salesforce orgs**.
 - Best for deploying metadata (not data).
-

6. Unmanaged vs Managed Packages

- Packages helped organize and share project components.
 - Salesforce **Packages** allow you to **bundle and distribute metadata** (custom objects, fields, flows, Lightning components, etc.) between orgs. This is especially useful when **Change Sets aren't available**.
-

1. Unmanaged Packages

Definition:

- A package that contains **customizations** you can install in another org.
- Once installed, the components are **editable** in the target org.

Use Case:

- Ideal for **Developer Edition projects** like your **Skill Development & Employment Portal**.

Pros:

- Easy to create and install.
- Editable in target org after installation.
- No Sandbox required.

Cons:

- No version control or upgrade path.
 - Cannot easily uninstall everything (components remain).
-

2. Managed Packages

Definition:

- A package created for **distribution via AppExchange**.
- Components are **locked/protected**, and updates can be pushed by the provider.

Use Case:

- Ideal for **commercial apps** that are distributed to multiple orgs.

Pros:

- Upgradeable.
- Components can be hidden or protected.
- Supports licensing and versioning.

Cons:

- Cannot edit core components in target org.
- More complex to create and maintain.

.Steps to Create an Unmanaged Package (Practical)

Step 1: Go to Package Manager

1. In **Setup** → **Package Manager**.
2. Click **New Package**.

Step 2: Enter Package Details

1. **Package Name:** e.g., `SkillDevPortal_Components`
2. **Description:** "Contains custom objects, fields, flows, and Lightning components for the Skill Development & Employment Portal project."
3. **Package Type:** Select **Unmanaged Package**
 - This allows you to **edit components in the target org** after installation.
4. Click **Save**

 **SETUP**

Package Manager

Create a Package

[Help for this Page](#) 

Package Edit

Save

Cancel

Package Name

Skill Development & Emplo

Language

English

Configure Custom Link

--None--

Notify on Apex Error

Description

Contains custom objects, fields, flows, and Lightning components for the Skill Development & Employment Portal project.

Managed

☐

Save

Cancel

Step 3: Add Components

1. After saving, click **Add Components**.
2. Select the component types you want to include:
 - **Custom Objects:** Candidate__c, Job__c, Skills__c
 - **Fields & Relationships** within these objects
 - **Validation Rules**
 - **Flows, Lightning Components, Record Types**
3. Add all required components → click **Add to Package**



SETUP

Package Manager

AgentPromotionEventSummary	☐	☐	☐	☐
AnalyticsUserAttrFuncTkn	☐	☐	☐	☐
ConversationApiLog	☐	☐	☐	☐
ConversationApiLogObjSum	☐	☐	☐	☐
DataObjSecondaryIndex	☐	☐	☒	☐
AiJobRun	☐	☐	☐	☐
AiJobRunItem	☐	☐	☐	☐
DataHarmonizedModelObjRef	☐	☐	☐	☐
WebCartCredit	☐	☐	☐	☐
AiGroundingSourceStage	☐	☐	☐	☐
AiGroundingWebRef	☐	☐	☐	☐
AppExchangeAnomalyEventStore	☐	☐	☐	☐
LoginAnomalyEventStore	☐	☐	☐	☐
ContactCenterBulkOp	☐	☐	☐	☐
Student__c	☐	☒	☒	☐
Skill__c	☐	☒	☒	☐
Application__c	☐	☒	☒	☐
enrollment__c	☐	☐	☒	☐

☒ = Auto-detected

Upload

Cancel

2. Select the **target org version** (optional).
3. Salesforce generates an **installation URL**.
4. Copy the URL → open it in the target org → click **Install**.

Step 5: Verify Installation

1. Log in to the **target org**.
2. Go to **Setup** → **Installed Packages**.
3. Check that **SkillDevPortal_Components** is installed and all objects/components are available.

◆ How we applied it in our project:

- **Unmanaged Package:** Shared our project internally for testing.
- **Managed Package (Future Scope):** To distribute the Skill Portal as a product for external organizations.

Key Points:

- **Unmanaged** → editable, good for development.
 - **Managed** → locked, versioned, suitable for AppExchange.
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7. ANT Migration Tool

The **ANT Migration Tool** was explored for CI/CD automation.

Key Points:

- Command-line tool.
 - Supports scripted deployments.
 - Useful for large projects with frequent changes.
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8. VS Code & SFDX

Salesforce DX (SFDX) with VS Code was used as our main development environment.

Step 1: Install Prerequisites

1. **Install VS Code** (already installed in your case).
2. **Install Salesforce CLI (SFDX)**
 - Download from Salesforce CLI.
 - After installation, confirm in terminal:

```
sfdx --version
```

```
C:\Users\ADMIN>sfdx --version
sfdx-cli/7.209.6 win32-x64 node-v18.15.0

C:\Users\ADMIN>|
```

3. Install Salesforce Extensions for VS Code

- Open VS Code → Extensions → Search **Salesforce Extension Pack** → Install.
We have already installed it in our earlier steps
-

◆ Step 2: Authorize Salesforce Org

1. Open VS Code → Terminal.
2. Run:

```
sfdx force:auth:web:login -d -a MySandbox
```

 - `-d` → sets as default org
 - `-a` → alias name (example: MySandbox)
3. A browser window opens → login with **Sandbox/Production** credentials.
4. After success → your org is now linked to VS Code.

Authentication Successful

You've successfully logged in. You can now
close this browser tab or window.

◆ Step 3: Create/Clone Project

- To create a new SFDX project:

```
sfdx force:project:create -n SkillPortal
```
- To open existing project → **File** → **Open Folder** → select your project.
- If already connected with GitHub (like your TCS_mile branch), just open that folder.

◆ Step 4: Retrieve Components from Org

Example: Retrieve all metadata (classes, lwc, objects, etc.):

```
sfdx force:source:retrieve -m ApexClass sfdx force:source:retrieve -m LightningComponentBundle sfdx force:source:retrieve -m CustomObject
```

Or full org metadata:

```
sfdx force:source:retrieve -u MySandbox -x manifest/package.xml
```

```
sfdx force:source:retrieve -m ApexClass
>> sfdx force:source:retrieve -m LightningComponentBundle
>> sfdx force:source:retrieve -m CustomObject
>>
```

SkillCertRESTCallout	ApexClass	force-app\main\default\classes\SkillCertRESTCallout.cls	Created
SkillCertRESTCallout	ApexClass	force-app\main\default\classes\SkillCertRESTCallout.cls-meta.xml	Created
SkillManager	ApexClass	force-app\main\default\classes\SkillManager.cls	Created
SkillManager	ApexClass	force-app\main\default\classes\SkillManager.cls-meta.xml	Created
SkillManagerSafe	ApexClass	force-app\main\default\classes\SkillManagerSafe.cls	Created
SkillManagerSafe	ApexClass	force-app\main\default\classes\SkillManagerSafe.cls-meta.xml	Created
SkillManagerSafeTest	ApexClass	force-app\main\default\classes\SkillManagerSafeTest.cls	Created
SkillManagerSafeTest	ApexClass	force-app\main\default\classes\SkillManagerSafeTest.cls-meta.xml	Created
TrainingHandler	ApexClass	force-app\main\default\classes\TrainingHandler.cls	Created
TrainingHandler	ApexClass	force-app\main\default\classes\TrainingHandler.cls-meta.xml	Created

Warning: We plan to deprecate this command in the future. Try using the "project retrieve start" command instead.
Retrieving v64.0 metadata from rupasree287375@ageantforce.com using the v64.0 SOAP API
Preparing retrieve request... done
=== Retrieved Source

FULL NAME	TYPE	STATE	PROJECT PATH
applicationTracker	LightningComponentBundle		force-app\main\default\lwc\applicationTracker\applicationTracker.html
applicationTracker	LightningComponentBundle		force-app\main\default\lwc\applicationTracker\applicationTracker.js
applicationTracker	LightningComponentBundle		force-app\main\default\lwc\applicationTracker\applicationTracker.js-meta.xml
candidateProfileCard	LightningComponentBundle		force-app\main\default\lwc\candidateProfileCard\candidateProfileCard.html

1. Open your project folder in **VS Code** (SkillDevEmploymentPortal).

2. In the terminal, make sure you're on the correct branch:

```
git checkout TCS_mile
```

3. Add all changes:

```
git add .
```

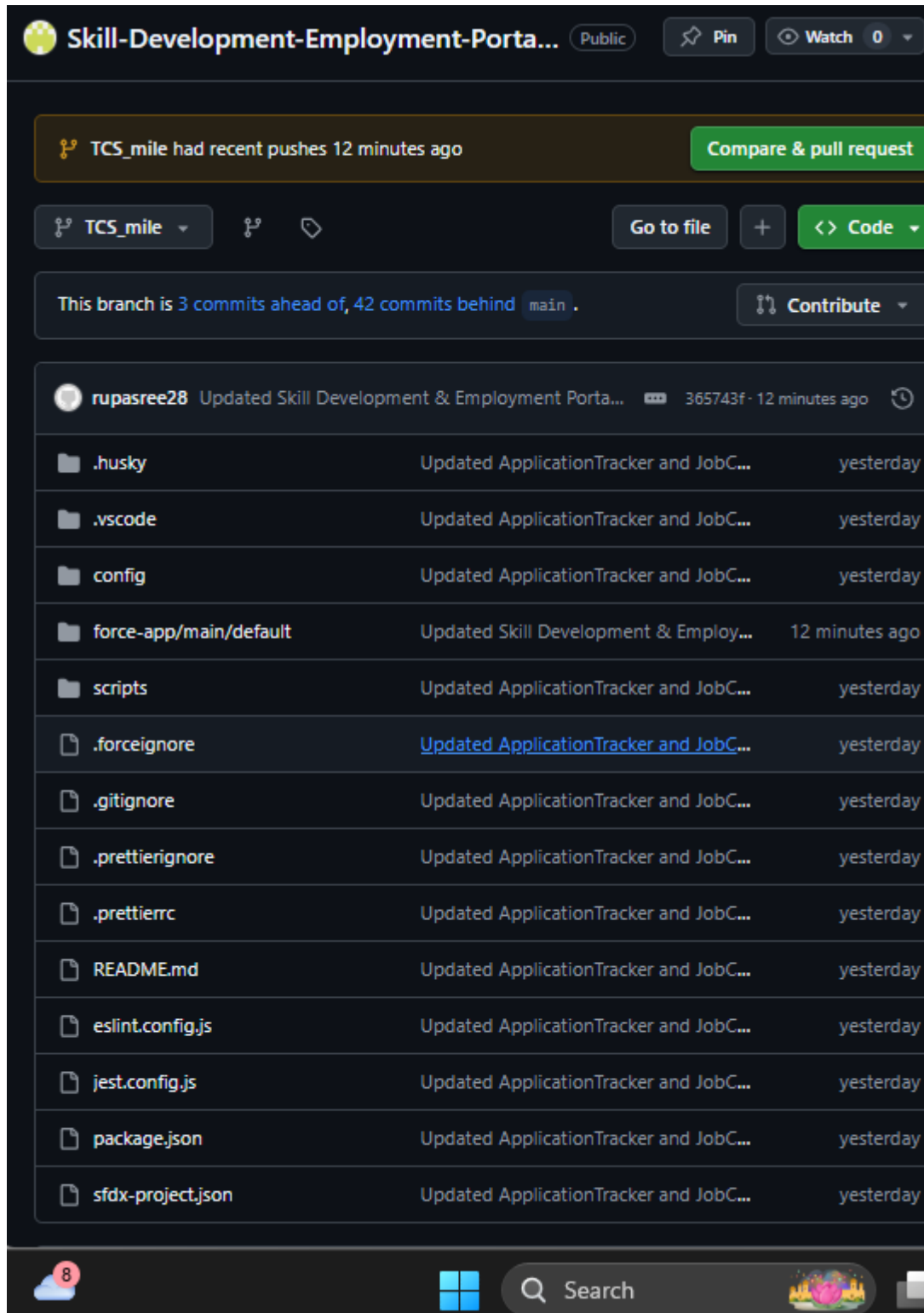
4. Commit your changes:

```
git commit -m "Updated Skill Development & Employment Portal with latest Salesforce changes"
```

5. Push to GitHub:

```
git push origin TCS_mile
```

↪ This phase ensures you can code, test, and sync your Salesforce project directly from VS Code, instead of relying only on the web UI. and changes had hit into github.



⚡ How we applied it in our project:

- Pulled metadata and pushed changes between Sandbox & VS Code.

- Used **Scratch Orgs** for isolated development and testing.
- Integrated with **GitHub (TCS_mile branch → future renamed)** for source control.
- Executed **SFDX commands** for deployment and retrieval.

Key Points:

- Developer-friendly CLI-based tool.
 - Encourages **source-driven development**.
 - Supports **continuous integration (CI/CD)** with pipelines.
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Outcomes of Phase 8

- Established **data integrity** with import/export, duplicate rules, and backups.
- Ensured **smooth deployment process** with Change Sets, ANT, and SFDX.
- Project is now **deployment-ready**, with both admin-friendly and developer-friendly tools available.
- Laid foundation for **scalable product distribution** via Managed Packages.